

CanoVerse: 3D Object Scalable Canonicalization and Dataset for Generation and Pose

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Outline

In Section 1, we provide additional methodological details, including data pre-processing, criterion design, the annotation procedure, and annotation screening statistics. Section 2 details the experimental settings for each task evaluated in Section 4 of the main paper. Section 3 presents additional visual results for downstream tasks and a screenshot of the annotation GUI. Section 4 shows more samples from our dataset.

1 Detailed Method Description

1.1 Data Preprocessing Details

Semantic text acquisition. For semantic part segmentation with Find3D [9], we require a per-category text prompt that contains a list of semantic parts for objects in that category. Following COD [4], we obtain this part list by querying large language models (e.g., GPT). Specifically, we ask for a concise set of part names describing the object structure (e.g., “wheel, frame, handlebar” for bicycles). This list is then used to guide part segmentation and subsequent semantic alignment.

Category template and cross-category harmonization. Each category is associated with a representative template whose orientation follows human convention (e.g., bicycle: z = front, y = up). For objects with a clear forward direction, such as cars and bicycles, the $+z$ axis is defined as the forward direction. For objects with ambiguous forward directions, such as swords and hammers, we define the functional end as the $+z$ direction and place the handle toward the $-z$ direction.

Cross-category harmonization is then achieved by manually aligning the orientations of these templates across all 1,156 categories (e.g., both cars and bicycles face the $+z$ direction). Since instances within each category are subsequently aligned to their corresponding template, ensuring consistent template orientations naturally enforces cross-category consistency across the entire dataset.

1.2 Vertical Alignment: Support Surface

Support-surface criterion. Following [4], we adopt a support-plane-based method for stable object pose initialization to address the issue of incorrect object upward orientation.

Given an input 3D point cloud $\mathcal{P} = \{\mathbf{p}_i \in \mathbb{R}^3 | i = 1, 2, \dots, N\}$ (where N denotes the number of points in the point cloud and $\mathbf{p}_i$ represents the 3D coordinates of the i -th point), we first compute its minimal 3D convex hull $\mathcal{CH}(\mathcal{P})$. The surface of this convex hull consists of M non-overlapping convex polygonal faces, denoted as the face set $\mathcal{F} = \{F_k | k = 1, 2, \dots, M\}$, with each convex polygonal face F_k serving as a candidate support face for stable placement.

For each candidate face F_k , our goal is to rotate the object accordingly such that this face becomes parallel to the ground plane. To achieve this, we compute a rotation that aligns the inward-pointing face normal $\mathbf{n}_s^k \in \mathbb{R}^3$ with the ground normal $\mathbf{n}_g \in \mathbb{R}^3$. The rotation axis $\mathbf{u}_k$ is defined as the cross product

$$\mathbf{u}_k = \mathbf{n}_s^k \times \mathbf{n}_g, \quad (1)$$

which is orthogonal to both normals and specifies the axis around which the face normal should rotate. The rotation angle $\theta_k \in [0, \pi]$ is computed as

$$\theta_k = \arccos \left(\frac{\mathbf{n}_s^k \cdot \mathbf{n}_g}{\|\mathbf{n}_s^k\| \|\mathbf{n}_g\|} \right), \quad (2)$$

which measures the angular difference between the two normals.

Given the axis-angle representation $(\mathbf{u}_k, \theta_k)$, we convert it into a 3×3 rotation matrix $\mathbf{R}_k$ using Rodrigues' rotation formula:

$$\mathbf{R}_k = \mathbf{I} + \sin \theta_k \mathbf{K}_k + (1 - \cos \theta_k) \mathbf{K}_k^2, \quad (3)$$

where $\mathbf{I}$ denotes the identity matrix and $\mathbf{K}_k$ is the skew-symmetric matrix corresponding to the axis $\mathbf{u}_k = (u_{k,x}, u_{k,y}, u_{k,z})^\top$. Finally, applying $\mathbf{R}_k$ to the original point cloud $\mathcal{P}$ yields the rotated point cloud

$$\mathcal{P}'_k = \{\mathbf{p}'_{i,k} = \mathbf{R}_k \mathbf{p}_i\}. \quad (4)$$

This rotation aligns the candidate face F_k parallel to the ground plane, producing a candidate pose for correcting the object's upward orientation.

To verify placement stability, we compute the centroid of the rotated point cloud $\mathbf{c}_k = \frac{1}{N} \sum_{i=1}^N \mathbf{p}'_{i,k}$ and project it onto the ground to obtain $\mathbf{c}_k^{\text{proj}}$. If $\mathbf{c}_k^{\text{proj}}$ lies within the boundary of the rotated face F'_k (the convex polygon obtained by transforming F_k with $\mathbf{R}_k$), the face is considered a valid support face, and the corresponding rotation matrix $\mathbf{R}_k$ yields a stable placement.

We render the original meshes under the poses obtained from the above process and use a VLM (e.g., ChatGPT) to select the pose that best matches human expectations of an "upright" orientation, denoted as $\mathbf{R}_{\text{Sup}}$.

1.3 Horizontal Alignment: Joint Semantic-Geometric Criterion

The semantic criterion combines a geometric energy $E_g(\theta)$ and a semantic energy $E_s(\theta)$ with a Gaussian prior over geometric extrema, which are given by:

$$E_g(\theta) = \text{CD}(\mathcal{P}^t, R_z(\theta) \mathcal{P}^o), \quad (5)$$

$$E_s(\theta) = \frac{1}{m} \sum_{k=1}^m \text{CD}(\mathcal{S}_k^t, R_z(\theta) \mathcal{S}_k^o), \quad (6)$$

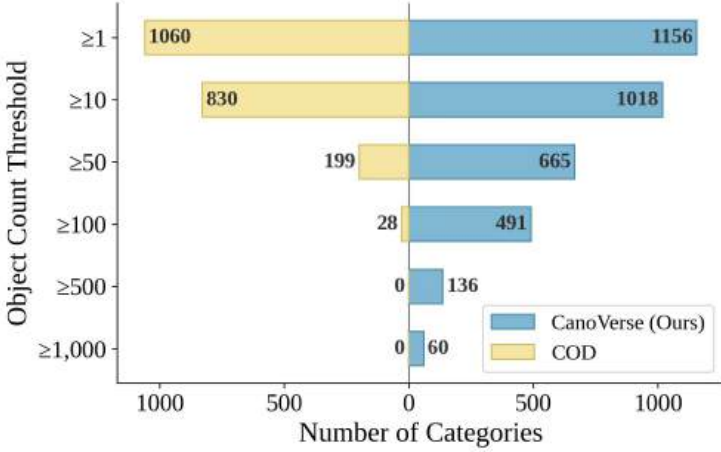


Fig. 1: Comparison of category counts under different instance thresholds between CanoVerse and COD [4]. CanoVerse provides substantially richer instance coverage per category, with many more categories containing hundreds of shapes.

where θ is the horizontal rotation angle around the vertical z -axis, $R_z(\theta)$ is the corresponding rotation matrix, $\mathcal{S}_k^o$ and $\mathcal{S}_k^t$ are the k -th semantic point sets of the object and template, $\mathcal{P}^o$ and $\mathcal{P}^t$ are the full point sets for the current instance and the template. Let $\Omega = \{\theta \mid \frac{\partial}{\partial \theta} E_g(\theta) = 0\}$ be the set of geometric extrema. The optimal angle is

$$\theta^* = \arg \min_{\theta \in [0, 2\pi)} - \left(e^{-E_s(\theta)} \cdot \sum_{\omega_k \in \Omega} \mathcal{N}(\theta \mid \omega_k, \mathbf{I}) \right), \quad (7)$$

where $\mathcal{N}(\theta \mid \omega_k, \mathbf{I})$ is a Gaussian centered at geometric extremum ω_k with identity covariance, favoring solutions that are both semantically consistent and near a geometric minimum. The semantic alignment matrix is $\mathbf{R}_S = R_z(\theta^*)$.

1.4 Annotation Protocol and Quality Control

Calibration and workforce. Each annotator completes a short calibration on 100 randomly selected objects before large-scale labeling; most annotators master the procedure within about 2 minutes of instruction. In total, 20 annotators performed parallel labeling, contributing roughly 900 working hours.

Quality control. We log per-category annotation time and inter-rater agreement, and run lightweight rechecking for low-confidence categories. Annotators are also instructed to discard instances that are misclassified or show severe geometric degradation (e.g., major mesh loss or incomplete structure).

Screening statistics. Across 749,736 annotated samples, 320K are retained. Of the discarded samples, 51% are removed due to low quality (e.g., thin-shell

meshes, meaningless structures) and 6% due to pose/alignment errors; the remaining 43% form the valid canonical set. The selection distribution over the five candidates (see Section 3.2 and Figure 5 in the main paper) indicates that all candidate types contribute to the final annotations. The final dataset exhibits a long-tailed category distribution, which is consistent with real-world object statistics and provides substantially more shapes than COD [4] in most categories. Figure 1 compares the number of categories under different instance-count thresholds. CanoVerse consistently contains far more categories with large numbers of shapes (e.g., ≥ 100 , ≥ 500 , and ≥ 1000 instances), whereas COD is dominated by sparse categories with fewer than 50 shapes. This indicates that CanoVerse provides substantially denser and more balanced category coverage.

2 Additional Experimental Details

2.1 3D Object Orientation Estimation

When training the object orientation estimation network, we uniformly sample 8192 points for each object, which are used as the point clouds for training. The orientation estimation task is trained on 8 96GB-VRAM NVIDIA GPUs.

During the training of VI-Net [6], AGPose [8], SecondPose [1], and CleanPose [7], we apply a random rotation to each point cloud before feeding it into the network, and use the corresponding rotation as the ground truth for loss calculation. During training, we apply 40 random rotations to each point cloud and train all networks for 40 epochs. For VI-Net and SecondPose, we use the Adam optimizer with an initial learning rate of $1e-3$, followed by a cosine annealing schedule. For CleanPose and AGPose, we use a learning rate ranging from $2e-5$ to $5e-4$ based on a triangular2 cyclical schedule. For CleanPose, AGPose, and SecondPose, we remove their RGB image input branches and retain only the point cloud input branches, as these networks originally require RGB images as input. Additionally, when training VI-Net on datasets of different scales, we use a total batch size of 1024 and train for 150k iterations.

2.2 3D Object Generation

Training. For fine-tuning both Hunyuan3D 2.1 [10] and Trellis [12], we use a subset of our canonicalized dataset containing 100K samples. The dataset is constructed by manually screening 100k objects with high aesthetic quality from the raw data. For Trellis [12], fine-tuning is performed for 100K steps with a total batch size of 40 and a learning rate of 1×10^{-5} . For Hunyuan3D 2.1 [10], we fine-tune for 20K steps with a total batch size of 42 and a learning rate of 5×10^{-6} . Fine-tuning Trellis on the canonicalized data is conducted on 5 NVIDIA GPUs, taking approximately 5 days. Fine-tuning Hunyuan3D 2.1 uses 7 NVIDIA GPUs and takes about 3 days. Fine-tuning with non-canonical data took the same amount of time as fine-tuning with canonical data for fair comparison. For all experiments, we use identical GPU cards, each with 96 GB of VRAM.

GT for pose stability evaluation. To evaluate the pose stability of generated 3D objects, we select 100 images from the dataset and generate corresponding 3D objects. We estimate the canonical ground-truth pose for each generated object using our canonicalization method, compute its pose error relative to the canonical space, and use the Interquartile Range (IQR) as the evaluation metric.

2.3 3D Shape Retrieval

For ULIP [13] and Uni3D [15], we uniformly sample 8,192 points per object and train using 8 96GB-VRAM GPUs. ULIP is trained with a batch size of 1,024 for 250 epochs, using a warm-up starting from 1×10^{-6} , a peak learning rate of 6×10^{-3} , and a final learning rate of 1×10^{-5} . Uni3D is trained with a total batch size of 128 for 20 epochs, using a learning rate of 1×10^{-3} with 10,000 warm-up steps.

2.4 Annotation Quality and Efficiency

This section provides more details in our experimental setup for evaluating different canonicalization methods, including the construction of evaluation samples and the definition of annotator proficiency levels.

Evaluation data. In the comparison with manual annotation, we randomly select 30 objects from the DREDS dataset [2] and apply random rotation perturbations before assigning them to annotators with different levels of experience. For the COD method [4], we apply its intermediate candidate poses to the objects and let annotators select the optimal pose.

Annotator proficiency levels. We evaluate all annotation methods using annotators of three distinct proficiency levels, defined as follows:

Low: fewer than 20 previously annotated objects; **Medium:** 20–40 previously annotated objects; **High:** more than 40 previously annotated objects.

3 Additional Experimental Results

To demonstrate the generalization ability of the orientation estimation network (VI-Net [6]) trained on our dataset, we evaluate it on real-world data (OmniObject3D [11]), synthetic data (Objaverse [3]), and generated objects. Before testing, we apply random rotations to the original meshes to avoid biases caused by potential canonical orientations in the datasets. See Figures 3, 4, and 5.

In addition to the orientation estimation task, training on canonical data also enables 3D generative models to generate 3D objects with consistent orientations and higher geometric quality. See Figure 6.



Fig. 2: Manual pose selection interface. Annotators select the image whose orientation best matches the template in the upper-left corner. If the category is inconsistent with the template or the quality is unacceptable, the Discard key is used. If no candidate shares the same orientation as the template, the Error key is selected.

4 Dataset Alignment Visualization

To illustrate the canonicalization quality of our dataset, we visualize samples from several common categories. As shown in Figure 7, our dataset achieves not only intra-class alignment but also inter-class alignment (Figure 7). For example, categories such as sharks and horses share a consistent orientation. Prior work has shown that inter-class mesh alignment improves the training of pose estimation networks [14].

References

1. Chen, Y., Di, Y., Zhai, G., Manhardt, F., Zhang, C., Zhang, R., Tombari, F., Navab, N., Busam, B.: Secondpose: Se (3)-consistent dual-stream feature fusion for category-level pose estimation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 9959–9969 (2024)
2. Dai, Q., Zhang, J., Li, Q., Wu, T., Dong, H., Liu, Z., Tan, P., Wang, H.: Domain randomization-enhanced depth simulation and restoration for perceiving and grasping specular and transparent objects. In: European Conference on Computer Vision (ECCV) (2022)
3. Deitke, M., Schwenk, D., Salvador, J., Weihs, L., Michel, O., VanderBilt, E., Schmidt, L., Ehsani, K., Kembhavi, A., Farhadi, A.: Objaverse: A universe of annotated 3d objects. arXiv preprint arXiv:2212.08051 (2022)
4. Jin, L., Wang, Y., Chen, W., Dai, Q., Gao, Q., Qin, X., Chen, B.: One-shot 3d object canonicalization based on geometric and semantic consistency. In: CVPR. pp. 16850–16859 (June 2025)

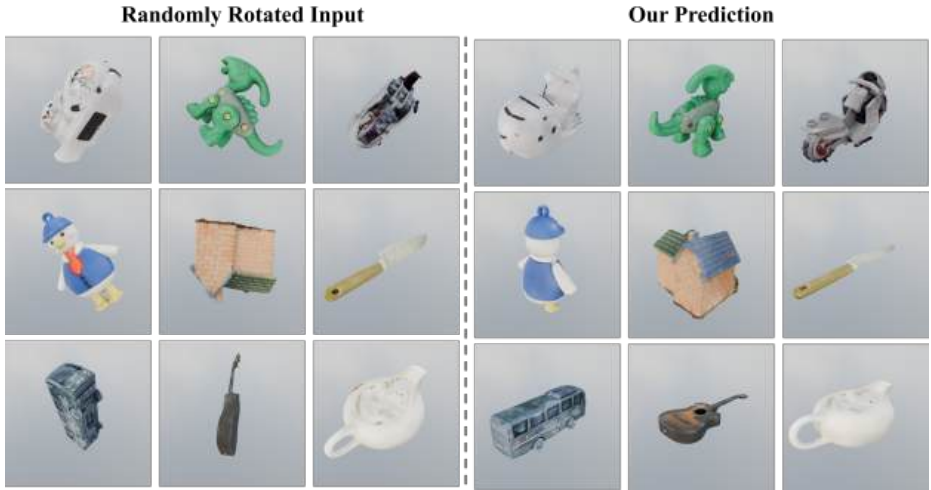


Fig. 3: The visualization results of zero shot orientation estimation on the real-world OmniObject3D [11] dataset. We used the scanned meshes provided by the dataset for rendering, while the point clouds from the dataset were adopted for inference.

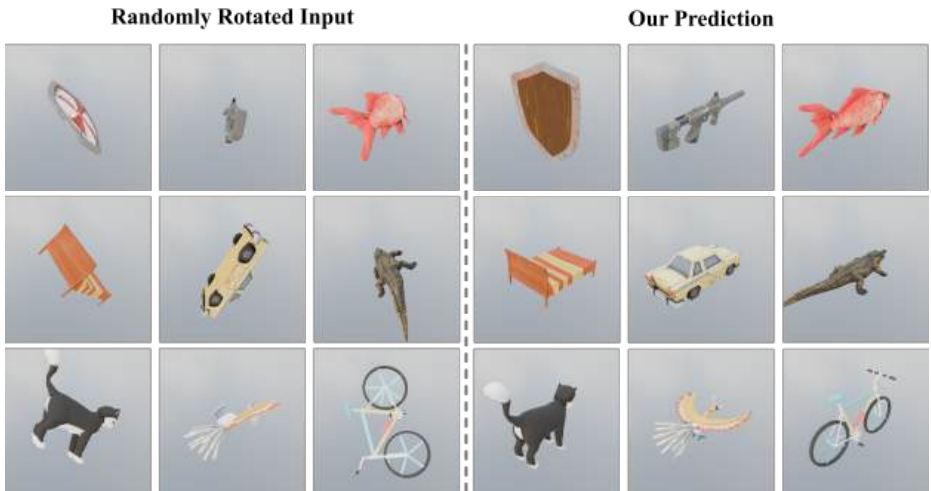


Fig. 4: The visualization results of zero shot orientation estimation on the Objaverse [3] dataset.

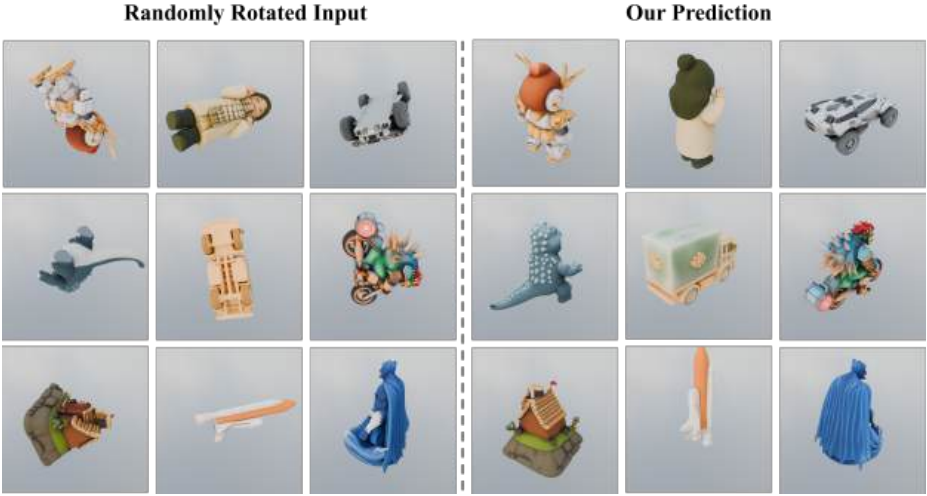


Fig. 5: The visualization results of zero shot orientation estimation on the meshes generated by Hunyuan3D 2.5 [5].

5. Lai, Z., Zhao, Y., Liu, H., Zhao, Z., Lin, Q., Shi, H., Yang, X., Yang, M., Yang, S., Feng, Y., et al.: Hunyuan3d 2.5: Towards high-fidelity 3d assets generation with ultimate details. arXiv preprint arXiv:2506.16504 (2025)

6. Lin, J., Wei, Z., Zhang, Y., Jia, K.: Vi-net: Boosting category-level 6d object pose estimation via learning decoupled rotations on the spherical representations. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 14001–14011 (2023)

7. Lin, X., Peng, Y., Wang, L., Zhong, X., Zhu, M., Yang, J., Feng, Y., Liu, C., Chen, Q.: Cleanpose: Category-level object pose estimation via causal learning and knowledge distillation. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) (2025)

8. Lin, X., Yang, W., Gao, Y., Zhang, T.: Instance-adaptive and geometric-aware keypoint learning for category-level 6d object pose estimation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 21040–21049 (2024)

9. Ma, Z., Yue, Y., Gkioxari, G.: Find any part in 3d (2024), <https://arxiv.org/abs/2411.13550>

10. Team, T.H.: Hunyuan3d 2.1: From images to high-fidelity 3d assets with production-ready pbr material (2025)

11. Wu, T., Zhang, J., Fu, X., Wang, Y., Jiawei Ren, L.P., Wu, W., Yang, L., Wang, J., Qian, C., Lin, D., Liu, Z.: Omniobject3d: Large-vocabulary 3d object dataset for realistic perception, reconstruction and generation. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2023)

12. Xiang, J., Lv, Z., Xu, S., Deng, Y., Wang, R., Zhang, B., Chen, D., Tong, X., Yang, J.: Structured 3d latents for scalable and versatile 3d generation. arXiv preprint arXiv:2412.01506 (2024)

13. Xue, L., Yu, N., Zhang, S., Panagopoulou, A., Li, J., Martín-Martín, R., Wu, J., Xiong, C., Xu, R., Niebles, J.C., et al.: Ulip-2: Towards scalable multimodal pre-

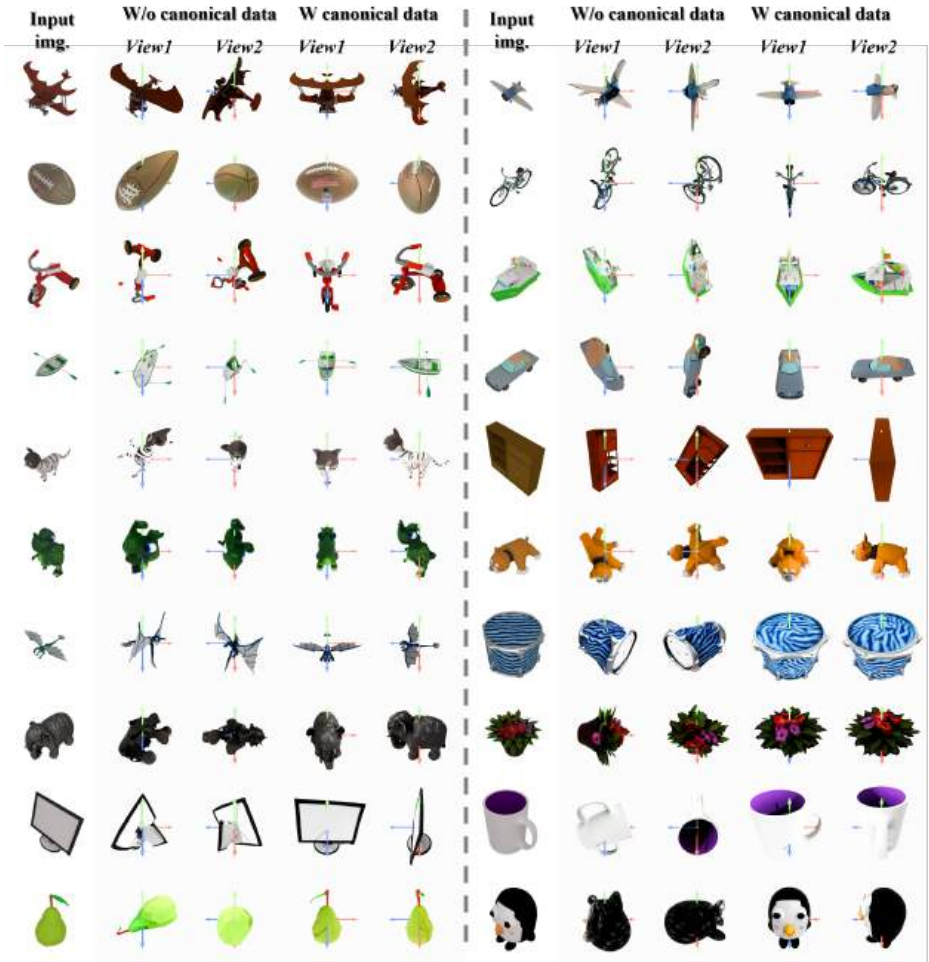


Fig. 6: Results for 3D object generation. Fine-tuning with canonical datasets ensures consistent object orientations, reduces semantic ambiguity, and enhances generation quality.



Fig. 7: Samples from the CanoVerse dataset. Our dataset achieves not only intra-class alignment but also inter-class alignment. For instance, the head orientations of horses and whales are consistent.

- training for 3d understanding. In: Proceedings of the IEEE/CVF Conference on
Computer Vision and Pattern Recognition. pp. 27091–27101 (2024)
14. Zhang, J., Huang, W., Peng, B., Wu, M., Hu, F., Chen, Z., Zhao, B., Dong, H.:
Omni6dpose: A benchmark and model for universal 6d object pose estimation and
tracking. In: European Conference on Computer Vision. pp. 199–216. Springer
(2024)
15. Zhou, J., Wang, J., Ma, B., Liu, Y.S., Huang, T., Wang, X.: Uni3d: Exploring
unified 3d representation at scale. In: International Conference on Learning Rep-
resentations (ICLR) (2024)